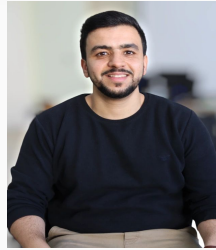


Data Mining Syllabus

Sami Belkacem & Mohammed Brahimi

Data Mining - Team

Dr. Sami Belkacem
Lecture & Lab instructor



Dr. Mohammed Brahimi
Lecture instructor



Dr. Meriem Amel Guessoum
Lab instructor



Dr. Oualid Ouarem
Lab instructor



Dr. Mohamed Akram Khelili
Lab instructor



Dr. Seif Eddine Bouziane
Lab instructor

Course Description

*“The Data Mining course focuses on **pattern extraction** from **data**, the purpose is to equip students with the capabilities to **manipulate data**, **preprocess it**, **extract patterns**, and **validate them**.”*

Learning Objectives

- **Master Core Data Mining Concepts**
 - Data preparation, classification, clustering, association rules, anomaly detection, etc.
- **Apply Data Mining in Practice**
 - Learn to develop and apply data mining techniques to real-world scenarios.
- **Enhance Analytical Thinking**
 - Learn to use data to rationalize and improve decision-making.
- **Prepare for Research and Application**
 - Gain skills for research in data mining or applying techniques to various fields.

Prerequisites

- **Statistics**

- Descriptive and inferential statistics.

- **Probability**

- Probability concepts.

- **Linear Algebra**

- Matrices and vectors.

- **Python Programming**

- Key for practical implementation.

Course content (1)

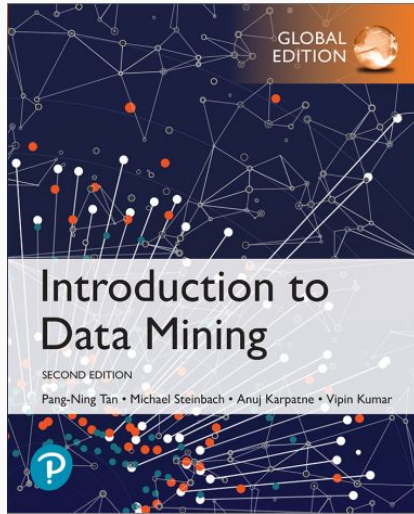
1	Introduction
2	Data (1)
3	Data (2)
4	Dimensionality Reduction (1)
5	Dimensionality Reduction (2)
6	Classification (1)
7	Classification (2)

Course content (2)

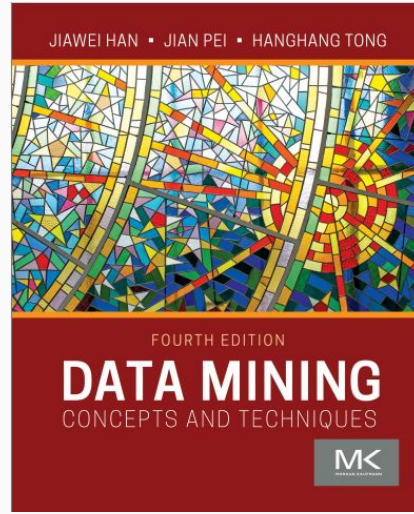
8	Clustering (1)
9	Clustering (2)
10	Association Analysis and Pattern Mining (1)
11	Association Analysis and Pattern Mining (2)
12	Anomaly Detection
13	False Discovery and Hypothesis Testing
14	Data Mining Use Cases

Textbooks

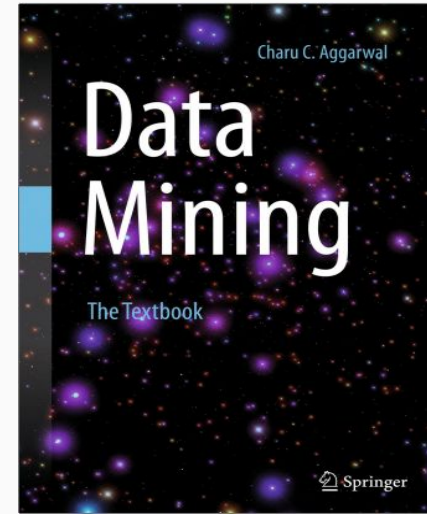
The main textbook for this course is the Textbook (1),
The Textbooks (2) and (3) are optional and only serve as a supplement to
enhance your understanding of the course material.



(1)



(2)



(3)

Assessment method & Grading policy

- Exam - 60%
- Continuous Evaluation - 40%
 - Midterm Exam - 6 points (**Week of midterms, November 8th**)
 - Data mining project - 6 points
 - Instructor appreciation - 4 points
 - Quizzes - 2 points
 - Attendance - 2 points (maximum of 2 absences)

Assessment method & Grading policy

- Late Quizzes
 - - **20%** penalty per day, up to 2 days
 - After 2 days, no points will be awarded
- Demonstration days will be organized for the Data Mining project

Attendance & Participation Policy

- Regular attendance is expected
- Participation in class discussions and group activities is much appreciated
- More than 3 absences without justification may result in disciplinary actions
- Unexcused lateness to class will not be accepted

Academic Honesty Policy

- All work must be original and completed to the best of ability
- **Plagiarism** and **cheating** will not be tolerated
- Appropriate disciplinary action will be taken for violations

"Education is not the filling of a pail, but the lighting of a fire."

التعليم هو إيقاد شعلة، وليس ملء وعاء – سقراط